WATER RESOURCES, WELLS & PUMPS · NOTE PU03

Centrifugal Pumps — Principle, Types, Parts, Characteristics, Installation, Priming and Troubles

Course topics: CAE1.3 L15, FMPE8.6 L6, IDE-511 T1, IDE-511 T2, IDE-511 T11

Why centrifugal pumps

They are the most common irrigation pumps: cheap, simple, easy to run and give a steady flow. Water is not positively enclosed — it flows continuously through the machine and pressure rises by dynamic action — so they are **non-positive** pumps.

Principle and main parts

- **Impeller (rotor):** a disc with vanes on the drive shaft that adds energy to the water as velocity and pressure.
- **Casing (volute):** a spiral whose cross-section grows towards the outlet; it guides water to and from the impeller.
- **Working:** the motor or engine spins the impeller fast; water enters at its centre (the **eye**), is caught by the vanes, accelerated and flung by centrifugal force into the volute and out of the discharge.
- The pump will not work until the casing is completely full of water — it must be **primed**.

Classification

Basis	Types
Energy conversion	Volute; diffuser
Number of stages	Single-stage; multi-stage
Impeller	Single or double suction; open, semi-open or closed
Axis of rotation	Horizontal; vertical
Drive	Direct-connected (coupled or close-coupled); geared; belt- or chain-driven

Type	Features
Volute	The most common: the impeller discharges into a gradually expanding spiral casing that slows the water and turns velocity head into pressure head
Diffuser	Stationary guide vanes round the impeller expand the passages and turn the flow, converting velocity head more completely before the volute — slightly more efficient, but complicated and costly, so used mainly in large high-pressure pumps; the vertical turbine pump has largely replaced it
Horizontal	Horizontal shaft, vertical impeller; the commonest irrigation pump — cheaper, easier to install and maintain, adaptable to any prime mover. Drawbacks: limited suction lift and liable to lose prime
Vertical	Vertical shaft with the impeller horizontal at the bottom; the casing can be lowered into the water so the impeller is submerged and priming is automatic; fits small wells and sumps with power at the surface; suits open wells with deep water columns (> 8 m) and rural areas with a deep water table and no electricity (engine at the surface)
Single- / multi-stage	One impeller develops the whole head, or two or more impellers in series (in one casing = multi-stage) each add energy; total head is proportional to the number of impellers
Single / double suction	Single: water enters one side — practical for small pumps but needs a large thrust bearing for the unbalanced axial thrust. Double: like two impellers back to back, water enters both sides, axially balanced — no thrust bearing needed

Impeller	Features
Open	Vanes only, no shroud; structurally weak (ribs or a partial shroud strengthen it); handles sediment and sand; low efficiency from slip
Semi-open	Shroud on one side; handles sediment-laden water; lower efficiency
Closed	Shrouds on both sides; most efficient; unsuited to sediment, which wears it
Non-clog	For sewage with much sediment, rags and solids: vanes well rounded at entry, large passages

Other parts

- **Materials:** cast-iron or bronze impellers for ordinary water; gun-metal for salt or brackish water; each impeller has a pair of wearing rings.
- **Casing:** usually cast iron, split horizontally (axially — suction and discharge nozzles in the lower half, upper half lifts off for inspection), vertically (radially — usually close-coupled designs) or diagonally.
- **Stuffing box** (gland and box) where the shaft leaves the casing: stops air leaking in when the pressure is below atmospheric and limits water leakage when above.
- **Bearings:** heavy-duty anti-friction ball bearings that carry thrust either way; grease occasionally. **Couplings:** flexible (most common, allow minor misalignment) or rigid (close-coupled pumps).
- **Bedplate:** mounts pump and motor or engine as one unit; level it, bolt it and grout it to a rigid foundation.
- **Foot valve** with strainer at the bottom of the suction pipe: holds water in the pump so it need not be primed each time and keeps debris out; needed in practically all horizontal centrifugal pumps.

Performance characteristics

Select a pump from the relationship of head, capacity, speed and efficiency, shown by characteristic curves plotted at constant speed: head–capacity, power–capacity and efficiency–capacity. Choose one with high efficiency over a wide range of discharge.

> Example from a manufacturer's curve (Fig. 6.29 of the textbook): efficiency ≥ 56% for 900–1400 L/min at heads of 27–20 m. The best operating point is 1110 L/min at 27 m head and 60% efficiency, needing 11 hp. Check: WHP = (1110/60 L/s × 27 m)/76 = 6.57 hp; shaft power = 6.57/0.60 ≈ 11 hp.

- Smooth, even flow; suited to high speeds. Capacity and head depend on rpm, impeller diameter and impeller width.
- At constant speed, capacity falls as head rises (and power falls); lowering the head or raising power increases capacity.

Change	Capacity	Head	Horsepower
Speed (N)	$\propto N$	$\propto N^2$	$\propto N^3$
Impeller diameter (D)	$\propto D$	$\propto D^2$	$\propto D^3$

Installation

1. **Locate** the pump for easy inspection, under cover (a pump house for permanent sets), safe from submergence and as close to the water as possible with a short, direct suction line.
2. **Foundation:** concrete, strong enough to absorb vibration, with bolts set in pipe sleeves about 2½ times the bolt diameter for adjustment. Level the unit on wedges (raising it about 1.5 cm), pour and cure the concrete, tighten the bolts and re-check alignment with shims.
3. **In wells**, fix the set on a platform of two steel beams with one end of each built into the well lining, leaving working space.
4. **Align the coupling** with a steel straight edge — no angular or parallel misalignment — and check the motor turns the way the arrow on the casing points.
5. **Piping:** let pipes line up naturally (never pull them in with the flange bolts) and support suction and delivery pipes independently.
6. **Suction line:** low friction; enough submergence to prevent vortices; no short elbow at the pump (sharp bends cause noise, loss of efficiency and heavy end thrust) — use long-radius bends. Prevent air entry from high intake velocity or whirlpools with a larger pipe (velocity below 1 m/s) or a mat or float over a shallow intake. Where there is much debris, put a screen 60–90 cm from the inlet; foot-valve strainers suffice for clear water.
7. **Delivery line:** fit a check (non-return) valve and a gate (regulating) valve when the lift is high or the pump fills an overhead tank — the check valve protects the pump from backflow surges; use the gate valve for priming and starting, and close it before stopping.
8. **Variable water levels:** use several platforms in the well; portable pumps on wheels or skids must stand level on firm, staked ground and suit only high water tables (pumping level within 6 m).

Suction-lift limits (horizontal pump)

- Up to 4.5 m for satisfactory operation; 6 m at most; efficiency drops sharply above about 4 m.
- Keep the pump above the water and as near it as possible. Electric sets can go down deep open wells (never submerge the motor); engine sets in deep wells waste power in long vertical belt drives.

- Little used in deep tube wells — deep pits are costly and drawdown usually exceeds 6 m. For streams, ponds and lakes, mount the pump on a float or provide for raising and lowering it.

Priming

- A non-positive pump lifts nothing until the casing and suction pipe are full of water.
- **Methods:** a foot valve with the casing and suction pipe filled from an overhead tank, the delivery pipe or a priming cup; or close the gate valve and exhaust the air with an air pump so water rises to replace it.

Operation and maintenance

- Never run the pump dry for more than a few seconds, or against a closed discharge (it overheats).
- **Every 6 months:** check the gland moves freely; clean and oil the gland bolts and nuts; inspect and if needed replace the packing; check alignment; drain and refill oil-lubricated bearings; check the grease in grease-lubricated ones.
- **Every year:** a thorough inspection — remove, clean and inspect the bearings and housings for flaws, scratches and wear, then re-oil or re-grease; remove the packing and check the shaft or sleeve for wear; disconnect the coupling halves and check alignment.

Troubles

Symptom	Likely causes
Fails to prime	Air leak in the suction line or pump; debris in the foot valve; shrunken gaskets; leaking rotary shaft seals; loose drain or fill plugs
Low pressure or capacity	Wrong speed; clogged suction line, strainer or foot valve; air leaks; excessive suction lift; worn wear rings or packing; impeller clogged
Takes too much power	Speed too high; head below the pump rating; bent shaft, binding rotating parts, tight stuffing box, misalignment
Leaks at the stuffing box	Packing worn or poorly lubricated; wrong or wrongly inserted packing
Noisy	**Cavitation** — excessive suction lift or high velocity drops the pressure below the vapour pressure and the vapour cavities collapse, damaging the impeller; or a bent shaft, worn or broken bearings, misalignment

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Lessons on pumps (classification, centrifugal and turbine pumps, pumping power). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. A centrifugal pump must be primed because it is

(a) a positive pump
(b) a non-positive pump
(c) self-priming
(d) submersible

Q2. For satisfactory operation, the suction lift of a horizontal centrifugal pump should not exceed about

(a) 2 m
(b) 4.5 m
(c) 8 m
(d) 10 m

Q3. If pump speed is doubled, the power required becomes

(a) 2 times (b) 4 times

(c) 8 times (d) unchanged

Q4. The impeller best suited to water carrying sand is the

(a) closed impeller (b) open impeller

(c) double-suction impeller (d) diffuser impeller

Q5. Noise from collapsing vapour bubbles in a pump is due to

(a) priming (b) cavitation

(c) water hammer (d) surging

Q6. The device at the bottom of the suction pipe that holds the prime is the

(a) gate valve (b) check valve

(c) foot valve (d) stuffing box

Answers

Q1 — (b) a non-positive pump

Q2 — (b) 4.5 m — *6 m is the maximum.*

Q3 — (c) 8 times — *Power* $\propto N^3$.

Q4 — (b) open impeller

Q5 — (b) cavitation

Q6 — (c) foot valve